

CONTACT

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CORE STACK

Python, SQL, JavaScript, Bash

TensorFlow, PyTorch, Scikit-learn, XGBoost

YOLOv8, OpenCV, DeepStream, TensorRT

Hugging Face, LLMs, RAG, FAISS

FastAPI, Flask, Django, Node.js

React, REST APIs, Responsive UI

Oracle, MySQL, MongoDB, SQLite

Docker, Nginx, Linux, GitHub Actions, CI/CD

EDUCATION

MS AI — FAST-NUCES, Islamabad
(2025 – Present)

BS AI — Hazara University, Mansehra
(2020 – 2024)

CERTIFICATIONS

Google IT Automation with Python Professional Certificate (2024)

Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp (2023)

SPECIALIZATIONS

Computer Vision & Video Analytics

Machine Learning & Deep Learning

Generative AI, LLMs & RAG

Full-Stack AI Applications

Anomaly & Fraud Detection

LANGUAGES

English — Professional working proficiency

Urdu — Native / bilingual proficiency

MUAVIA SHAKEEL

AI/ML ENGINEER | FULL-STACK AI ENGINEER

PROFESSIONAL PROFILE

Production-focused AI/ML Engineer and full-stack developer experienced in architecting real-time computer vision systems, enterprise automation, and AI-enabled web applications. Builds end-to-end solutions across model development, Python and Node.js backends, React dashboards, Oracle databases, and production deployment with Docker, Nginx, and CI/CD.

PROFESSIONAL EXPERIENCE

AI/ML Engineer
Aug 2025 – Present

Kohinoor Textile Mills Limited (KTML) — Rawalpindi, Pakistan | On-site

- Architected and deployed an end-to-end Visual AI Surveillance Platform across 40 live RTSP cameras, supporting people counting, speed-violation detection, facial recognition, vehicle classification, and ANPR.
- Engineered two concurrent DeepStream 7.1 inference pipelines on an RTX 5090 using YOLOv8, TensorRT FP16/INT8, and OpenCV, delivering approximately 120 FPS with sub-500 ms processing latency.
- Built FAISS-based facial recognition for 10,000 employee embeddings and a resilient camera layer with database-driven configuration, auto-reconnect, freeze detection, and headless operation.
- Developed the React and FastAPI dashboard for live views, alerts, counts, ANPR records, speed violations, face logs, analytics, filters, and reports; used by 3-5 Administrative Department users.
- Integrated Oracle event logging and evidence snapshots, then containerized services with Docker and Nginx. Production system live at one mill site since Nov 2025.
- Built and launched the Yarn ASM WhatsApp Bot in Mar 2026 for five sales users, enabling remote yarn booking, status tracking, stock checks, reports, and approvals through Meta WhatsApp Cloud API, n8n, FastAPI, and Oracle Database.
- Removed office and laptop dependency for booking workflows while reducing manual reporting and errors and improving access to sales information.

Management Trainee Officer (MTO)
Feb 2025 – Aug 2025

Kohinoor Textile Mills Limited (KTML) — Rawalpindi, Pakistan | On-site

- Built a production CNIC OCR and scanner application using Python, A8Capture.dll, Google Vision API, Urdu-to-English translation, automatic CNIC formatting, and Windows executable packaging.
- Deployed the solution across KTM and KGM in Jul 2025; processes approximately 100–150 CNICs per day in 1–3 seconds each with about 99% accuracy, reducing manual entry and improving speed and data quality.

Research Assistant — AI/ML Projects
Oct 2023 – Jan 2025

Hazara University — Mansehra, Pakistan | Hybrid

- Supported supervisor-led research by implementing selected AI/ML topics, running experiments, preparing datasets, and building reproducible development prototypes.
- Developed and optimized models for anomaly detection, predictive analysis, and classification, with selected experiments reaching 95%+ accuracy.
- Automated data-preprocessing pipelines, reducing processing time by 50%, and deployed proof-of-concept services through Flask and FastAPI.
- Explored GPT, LLaMA, and RAG workflows for document retrieval and chatbot development.

Python Developer Intern
Jun 2023 – Aug 2023

Kohinoor Textile Mills Limited — Rawalpindi, Pakistan | On-site

- Developed Python automation scripts for production-data analysis, improving workflow efficiency by 30%.
- Integrated SQL-based inventory tracking and reporting, contributed to Flask/FastAPI REST APIs, and built an operational analytics dashboard.

ACADEMIC RESEARCH & FINAL YEAR PROJECT

MS Thesis / Research — Federated Learning and Explainable Medical AI
Aug 2025 – Present

- Authored the manuscript "Federated Learning with Dataset-Size-Weighted GradCAM Aggregation for Explainable Multi-Label Chest X-Ray Diagnosis" and submitted it to Springer's Machine Learning journal.
- Proposed a privacy-preserving, dataset-size-weighted GradCAM aggregation method that combines client-level saliency maps without sharing raw medical images.
- Achieved an AUC-ROC of 0.8235 on NIH ChestX-ray14, covering 112,120 images and 14 pathology classes across five federated clients.
- Implemented and statistically validated an end-to-end pipeline using FedAvg, FedProx, Dirichlet non-IID partitioning, GradCAM faithfulness evaluation, and DP-FedAvg across multiple seeds and heterogeneity levels.

BS Final Year Project — The Malware Hunter: Applying Machine Learning for Cybersecurity
Feb 2024 – Jul 2024

- Developed a machine-learning-based malware detection system that achieved 90% classification accuracy and reduced false positives by 20%.
- Trained and evaluated XGBoost and Scikit-learn models for fast and reliable threat classification.
- Built a web-based detection interface and deployed the solution using FastAPI, Flask, and Streamlit for real-time malware analysis.